Assingment—7

Task Description #1 (Syntax Errors – Missing Parentheses in Print Statement)

Bug code:

```
name="shyam"

print(name
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
  File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 2
    print(name
          ^
SyntaxError: '(' was never closed
PS C:\Users\prath\.vscode\github>
```

prompt: find the error and fix the error and give correct code.

#The error in the code is that the print statement is not properly closed with a parenthesis. Here is the corrected code:

Correct code:

```
name = "shyam"

print(name)
```

```
n=10

  print("n is equal to 10")

else:

  print("n is not equal to 10")
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
prathap
PS C:\Users\prath\.vscode\github>
```

Task Description #2 (Incorrect condition in an If Statement)

Bug code:

```
n=10

if n=10:

   print("n is equal to 10")

else:

   print("n is not equal to 10")
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
  File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 2
    if n=10:
       ^^^^
SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?
```

Prompt: find the error and fix the error and give correct code.

Correct code:

```
n = 10

if n == 10:

   print("n is equal to 10")

else:

   print("n is not equal to 10")
```

Output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
n is equal to 10
PS C:\Users\prath\.vscode\github>
```

Task Description #3 (Runtime Error – File Not Found)

Bug code:

def read_file(filename):

with open(filename, 'r') as f:

return f.read()

print(read_file("nonexistent.txt"))

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
  File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 2
    with open(filename, &#39;r&#39;) as f:
    ^^^^
IndentationError: expected an indented block after function definition on line 1
```

prompt: solve the error and give the corect code and explain that

correct code:

#The error in the provided code occurs because the file "nonexistent.txt" does not exist, which raises a FileNotFoundError. To handle this error gracefully, we can use a try-except block to catch the exception and provide a meaningful message instead of crashing the program.

#Here is the corrected code with error handling:

def read_file(filename):

    try:

```
    with open(filename, 'r') as f:
        return f.read()
  except FileNotFoundError:
    return f"Error: The file '{filename}' does not exist."
print(read_file("da.txt"))
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
how are you.
PS C:\Users\prath\.vscode\github>
```

Task Description #4 (Calling a Non-Existent Method)

Task: Give a class where a non-existent method is called (e.g., obj.undefined_method()). Use AI to debug and fix.

Bug code:

```
class Car:
def start(self):
return &amp;quot;Car started&amp;quot;
my_car = Car()
print(my_car.drive())
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
  File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 2
    def start(self):
    ^^^
IndentationError: expected an indented block after class definition on line 1
```

prompt: find the error and fix the erroe and give correct code.

#The error in the code is that the method 'start' is not properly indented and the method 'drive' is called instead of 'start'. Here is the corrected code:

Correct code:

```
class Car:
```

```
    def start(self):
        return "Car started"

my_car = Car()
print(my_car.start())
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
Car started
PS C:\Users\prath\.vscode\github>
```

Task Description #5 (TypeError – Mixing Strings and Integers in Addition)

Task: Provide code that adds an integer and string ("5" + 2) causing a TypeError.

Bug code:

```
x="5"
y=2
print(x+y)
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
Traceback (most recent call last):
  File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 3, in <module>
    print(x+y)
          ~^~
TypeError: can only concatenate str (not "int") to str
```

prompt: find the error and fix the error and give correct code.

#The error in the code is that 'x' is a string and 'y' is an integer, which causes a TypeError when trying to add them. To fix this, we can convert 'x' to an integer before adding. Here is the corrected code:

Correct code:

```
x = "5"

y = 2

print(int(x) + y)
```

output:

